

NETWORK INFRASTRUCTURE DOCUMENTATION

Enterprise Network Addressing Plan



Dual-Site Enterprise Architecture — Headquarters & Branch Office

Dual-ISP SD-WAN Design with Site-to-Site IPsec VPN

PROJECT TITLE

Hybrid Enterprise Network Infrastructure — Final Year Project

AUTHOR

ELMahdi ZAHIR

DOCUMENT VERSION

1.0

DATE

July 2026

CLASSIFICATION

Project Documentation

Table of Contents

EnterpriseNetworkAddressingPlan—DocumentStructure

01	Document Information	3
02	Network Overview	4
03	WAN Addressing Plan	5
04	Headquarters Addressing	7
05	Branch Addressing	8
06	Headquarters VLAN Addressing Plan	9
07	Branch VLAN Addressing Plan	10
08	Transit Networks	11
09	Internal Network Summary	12
10	SD-WAN Members	13
11	VPN Protected Networks	14
12	Addressing Summary	15

About this document. This addressing plan defines the complete IPv4 numbering scheme for a dual-site enterprise network lab, covering WAN transit links, VLAN segmentation, SD-WAN overlay members, and IPsec VPN protected networks for both the Headquarters and Branch sites.

01

Document Information

Project Title	Hybrid Enterprise Network Infrastructure — Final Year Project
Author	ELMahdi ZAHIR
Version	1.0
Date	July 2026
Purpose	To define and document the complete IPv4 addressing plan for a dual-site enterprise network — including WAN transit links, dual-ISP connectivity, SD-WAN overlay, site-to-site IPsec VPN, and VLAN segmentation — for deployment reference, configuration consistency, and inclusion as supporting technical documentation in the associated final year project repository.

Scope. This document covers Layer 3 addressing only (WAN, transit, and VLAN subnetting). Active Directory, DNS, DHCP scopes, and cloud identity services are addressed in separate project documentation and are out of scope here.

02

Network Overview

The network consists of two sites — Headquarters (HQ) and a Branch office — interconnected over a redundant dual-ISP WAN using FortiGate next-generation firewalls with SD-WAN. A simulated ISP core (Core ISP router, ISP1, ISP2) provides transit between sites, and a site-to-site IPsec VPN tunnel with two redundant members protects inter-site traffic. Each site implements a three-tier switching hierarchy with VLAN-based segmentation for Servers, IT, HR, Sales, and Management traffic.

2

SITES (HQ &
BRANCH)

2

ISPS PER SITE
(DUAL-WAN)

6

VLANS PER SITE

2

REDUNDANT IPSEC
TUNNELS

Architecture Summary

- **Core ISP Router** — simulates the internet core, providing transit between both upstream ISPs (ISP1, ISP2) and both sites.
- **ISP1 / ISP2 Routers** — represent independent transit providers, each terminating a WAN link at HQ and at the Branch.
- **FortiGate HQ / FortiGate Branch** — perimeter firewalls providing dual-WAN SD-WAN, firewall policy enforcement, and IPsec VPN termination.
- **HQ Core Switch / Branch L3 Switch** — Layer 3 distribution point routing between VLANs and the FortiGate transit link.
- **VLAN Segmentation** — identical department structure (Servers, IT, HR, Sales, Management, Native) replicated at both sites using distinct summarized address blocks.

Design principle. HQ and Branch use a consistent VLAN numbering scheme (VLANs 10/20/30/40/99/999) mapped to distinct /16 address spaces (10.10.0.0/16 for HQ and 10.20.0.0/16 for Branch), simplifying firewall policy design, VPN route summarization, and troubleshooting.

03

WAN Addressing Plan

3.1 — Core ISP Router

INTERFACE	CONNECTED TO	NETWORK	IP ADDRESS
Ethernet0/1	ISP1-HQ	172.16.1.0/30	172.16.1.1
Ethernet0/2	ISP2-HQ	172.16.2.0/30	172.16.2.1
Ethernet0/3	ISP1-BR	172.16.3.0/30	172.16.3.1
Ethernet1/0	ISP2-BR	172.16.4.0/30	172.16.4.1

3.2 — ISP1 Router

INTERFACE	CONNECTED TO	NETWORK	IP ADDRESS
e0/0	Core ISP	172.16.1.0/30	172.16.1.2
e0/1	HQ FortiGate WAN1	100.64.1.0/24	100.64.1.1
e0/2	Branch FortiGate WAN1	100.64.3.0/24	100.64.3.1

3.3 — ISP2 Router

INTERFACE	CONNECTED TO	NETWORK	IP ADDRESS
e0/0	Core ISP	172.16.2.0/30	172.16.2.2
e0/2	Core ISP	172.16.4.0/30	172.16.4.2

INTERFACE	CONNECTED TO	NETWORK	IP ADDRESS
e0/1	HQ FortiGate WAN2	100.64.2.0/24	100.64.2.1
e0/3	Branch FortiGate WAN2	100.64.4.0/24	100.64.4.1

04

Headquarters Addressing

HQ

FortiGate HQ

INTERFACE	ROLE	IP ADDRESS
port1	WAN1	100.64.1.2/24
port2	WAN2	100.64.2.2/24
port3	HQ Core Switch	10.255.255.1/30

HQ Core Switch

INTERFACE	CONNECTED TO	IP ADDRESS
Transit	FortiGate HQ	10.255.255.2/30

Default Gateway: 10.255.255.1

05

Branch Addressing

BR

FortiGate Branch

INTERFACE	ROLE	IP ADDRESS
port1	WAN1	100.64.3.2/24
port2	WAN2	100.64.4.2/24
port3	Branch Layer 3 Switch	10.255.254.1/30

Branch Layer 3 Switch

INTERFACE	CONNECTED TO	IP ADDRESS
Transit	FortiGate Branch	10.255.254.2/30

Default Gateway: 10.255.254.1

06

Headquarters VLAN Addressing Plan

VLAN	DEPARTMENT	NETWORK	GATEWAY
10	Servers	10.10.10.0/24	10.10.10.1
20	IT	10.10.20.0/24	10.10.20.1
30	HR	10.10.30.0/24	10.10.30.1
40	Sales	10.10.40.0/24	10.10.40.1
99	Management	10.10.99.0/24	10.10.99.1
999	Native VLAN	No IP	—

Summary block: All Headquarters VLANs fall within 10.10.0.0/16, allowing a single summarized route to be advertised toward the Branch site over the VPN.

07

Branch VLAN Addressing Plan

VLAN	DEPARTMENT	NETWORK	GATEWAY
10	Servers	10.20.10.0/24	10.20.10.1
20	IT	10.20.20.0/24	10.20.20.1
30	HR	10.20.30.0/24	10.20.30.1
40	Sales	10.20.40.0/24	10.20.40.1
99	Management	10.20.99.0/24	10.20.99.1
999	Native VLAN	No IP	—

Summary block: All Branch VLANs fall within **10.20.0.0/16**, mirroring the HQ VLAN structure while remaining independently summarizable for VPN and firewall policy purposes.

08

Transit Networks

NETWORK	PURPOSE
172.16.1.0/30	Core ISP ↔ ISP1 HQ
172.16.2.0/30	Core ISP ↔ ISP2 HQ
172.16.3.0/30	Core ISP ↔ ISP1 Branch
172.16.4.0/30	Core ISP ↔ ISP2 Branch
100.64.1.0/24	ISP1 ↔ HQ FortiGate
100.64.2.0/24	ISP2 ↔ HQ FortiGate
100.64.3.0/24	ISP1 ↔ Branch FortiGate
100.64.4.0/24	ISP2 ↔ Branch FortiGate
10.255.255.0/30	HQ FortiGate ↔ HQ Core Switch
10.255.254.0/30	Branch FortiGate ↔ Branch Layer 3 Switch

Note. The 100.64.0.0/10 range (Shared Address Space, RFC 6598) is used here to simulate carrier-grade NAT segments between the ISP routers and each site's FortiGate WAN interfaces, consistent with real-world ISP hand-off addressing.

09

Internal Network Summary

SITE	ADDRESS SPACE
Headquarters	10.10.0.0/16
Branch	10.20.0.0/16

Each site's internal address space is a single summarizable /16 block, containing all VLAN subnets for that site. This design keeps route tables compact across the WAN and VPN, and simplifies firewall policy and NAT rule definition on both FortiGate appliances.

10

SD-WAN Members

Headquarters

HQ

PRIORITY	INTERFACE
1	WAN1 (port1)
2	WAN2 (port2)
10	HQ-BR-WAN1
20	HQ-BR-WAN2

Branch

BR

PRIORITY	INTERFACE
1	WAN1 (port1)
2	WAN2 (port2)
10	BR-HQ-WAN1
20	BR-HQ-WAN2

Priority logic. Local underlay WAN interfaces (priority 1–2) are preferred for general internet-bound traffic, while the SD-WAN overlay members (priority 10–20) — built over the site-to-site IPsec tunnels — carry inter-site traffic between HQ and Branch.

11

VPN Protected Networks

TUNNEL	PROTECTED NETWORK
HQ → Branch	10.20.0.0/16
Branch → HQ	10.10.0.0/16

The site-to-site IPsec VPN uses summarized /16 selectors on each side, so that all present and future VLANs within a site are automatically covered by the tunnel's Phase 2 selectors without requiring per-VLAN VPN entries.

12

Addressing Summary

Address Allocation Summary

BLOCK	ALLOCATION
172.16.0.0/16 (subset)	ISP core transit links (Core ISP ↔ ISP1/ISP2)
100.64.0.0/10 (subset)	ISP-to-FortiGate WAN hand-off links (RFC 6598 Shared Address Space)
10.255.254.0/30 – 10.255.255.0/30	FortiGate ↔ Core/L3 Switch transit links
10.10.0.0/16	Headquarters internal VLANs
10.20.0.0/16	Branch internal VLANs

Network Design Notes

- Every WAN and transit segment uses a /30 point-to-point network, minimizing address waste on router-to-router links.
- Each site's VLANs are contained within a single /16, enabling clean route summarization toward the WAN and VPN.
- Identical VLAN IDs are reused across sites (10, 20, 30, 40, 99, 999) with distinct third octets, keeping switch and firewall configuration templates consistent between HQ and Branch.
- Dual-ISP, dual-WAN design at both sites enables SD-WAN failover and load-balancing, with two independent IPsec tunnel members providing VPN redundancy.
- VLAN 999 is reserved as the native VLAN on all trunk links and is intentionally left unnumbered, per standard trunk-hardening practice.

IP Addressing Conventions Used

- **.1** – reserved for the default gateway on every VLAN and transit subnet.
- **.2** – assigned to the second device on point-to-point (/30) transit links.
- **Third octet = VLAN ID convention** – e.g. VLAN 10 → x.x.10.0/24, VLAN 20 → x.x.20.0/24, for fast visual correlation between VLAN ID and subnet.
- **Second octet = site convention** – 10.10.0.0/16 for HQ, 10.20.0.0/16 for Branch.

Reserved Address Ranges

RANGE	RESERVATION
10.10.50.0/24 – 10.10.98.0/24	Reserved for future HQ VLAN expansion
10.20.50.0/24 – 10.20.98.0/24	Reserved for future Branch VLAN expansion
10.10.100.0/16 – 10.10.255.0/24	Reserved / unallocated within the HQ block

RANGE	RESERVATION
10.20.100.0/16 – 10.20.255.0/24	Reserved / unallocated within the Branch block
172.16.5.0/30 – 172.16.255.0/30	Reserved for additional ISP transit expansion

Acronyms

WAN	Wide Area Network
LAN	Local Area Network
VLAN	Virtual Local Area Network
ISP	Internet Service Provider
SD-WAN	Software-Defined Wide Area Network
VPN	Virtual Private Network
IPsec	Internet Protocol Security
IKEv2	Internet Key Exchange version 2
HQ	Headquarters
BR	Branch
L3	Layer 3 (Network Layer)
RFC	Request for Comments
NAT	Network Address Translation
DHCP	Dynamic Host Configuration Protocol